



PUC Minas

Data Mesh

Marco Teixeira

O que vamos abordar nesta aula

O que é Data Mesh

O que são domínios de dados

Produtos de dados

Plataforma de dados

Governança Federada

O que vamos abordar nesta aula

O que é Data Mesh

O que são domínios de dados

Produtos de dados

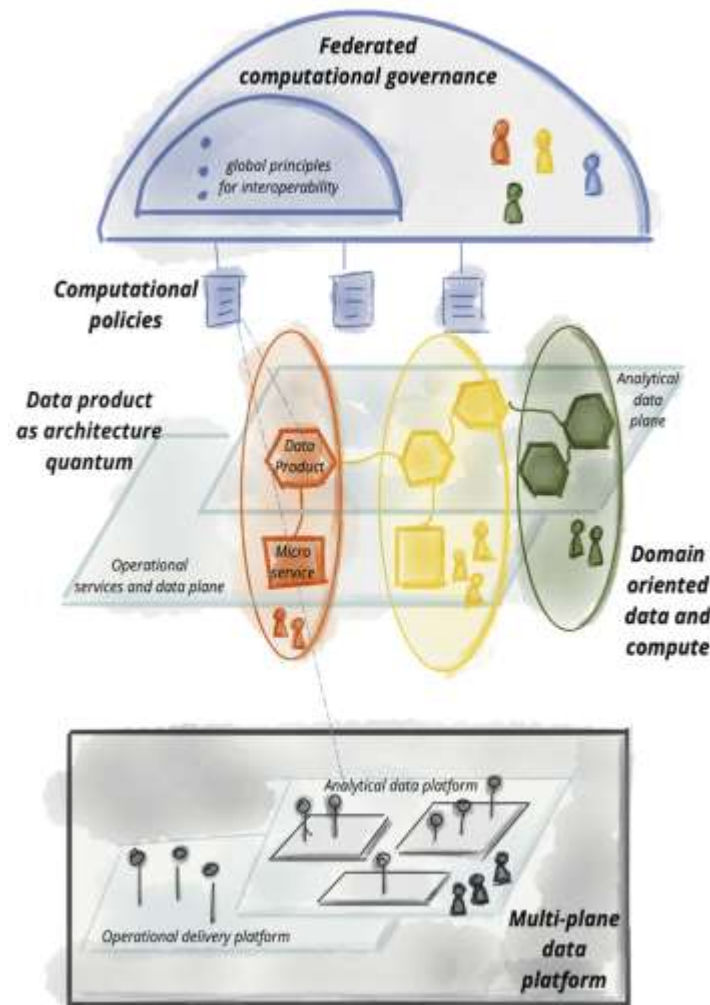
Plataforma de dados

Governança Federada

O que é o Data Mesh



Zhamak Dehghani



Domain-oriented decentralized data ownership and architecture

So that the ecosystem creating and consuming data can scale out as the number of sources of data, number of use cases, and diversity of access models to the data increases; simply increase the autonomous nodes on the mesh.

Data as a product

So that data users can easily discover, understand and securely use high quality data with a delightful experience; data that is distributed across many domains.

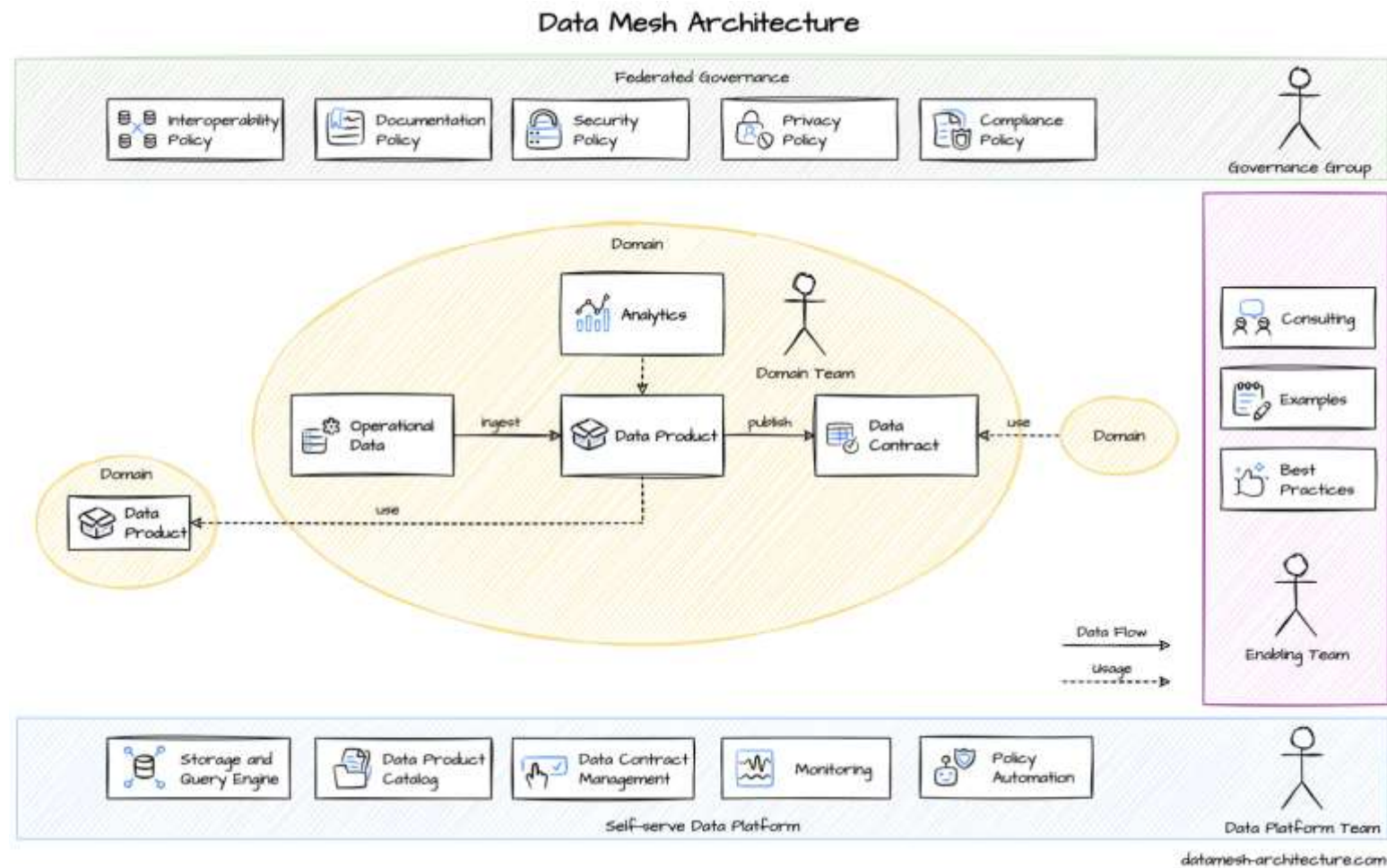
Self-serve data infrastructure as a platform

So that the domain teams can create and consume data products autonomously using the platform abstractions, hiding the complexity of building, executing and maintaining secure and interoperable data products.

Federated computational governance

So that data users can get value from aggregation and correlation of independent data products - the mesh is behaving as an ecosystem following global interoperability standards; standards that are baked computationally into the platform.

Data Mesh: Visão Global



Data Mesh: Princípios

Domain-oriented decentralized data ownership and architecture	So that the ecosystem creating and consuming data can scale out as the number of sources of data, number of use cases, and diversity of access models to the data increases; simply increase the autonomous nodes on the mesh.
Data as a product	So that data users can easily discover, understand and securely use high quality data with a delightful experience; data that is distributed across many domains.
Self-serve data infrastructure as a platform	So that the domain teams can create and consume data products autonomously using the platform abstractions, hiding the complexity of building, executing and maintaining secure and interoperable data products.
Federated computational governance	So that data users can get value from aggregation and correlation of independent data products - the mesh is behaving as an ecosystem following global interoperability standards; standards that are baked computationally into the platform.

Data Mesh: Princípios

Domain-oriented decentralized data ownership and architecture	So that the ecosystem creating and consuming data can scale out as the number of sources of data, number of use cases, and diversity of access models to the data increases; simply increase the autonomous nodes on the mesh.
Data as a product	So that data users can easily discover, understand and securely use high quality data with a delightful experience; data that is distributed across many domains.
Self-serve data infrastructure as a platform	So that the domain teams can create and consume data products autonomously using the platform abstractions, hiding the complexity of building, executing and maintaining secure and interoperable data products.
Federated computational governance	So that data users can get value from aggregation and correlation of independent data products - the mesh is behaving as an ecosystem following global interoperability standards; standards that are baked computationally into the platform.

O que vamos abordar nesta aula

O que é Data Mesh

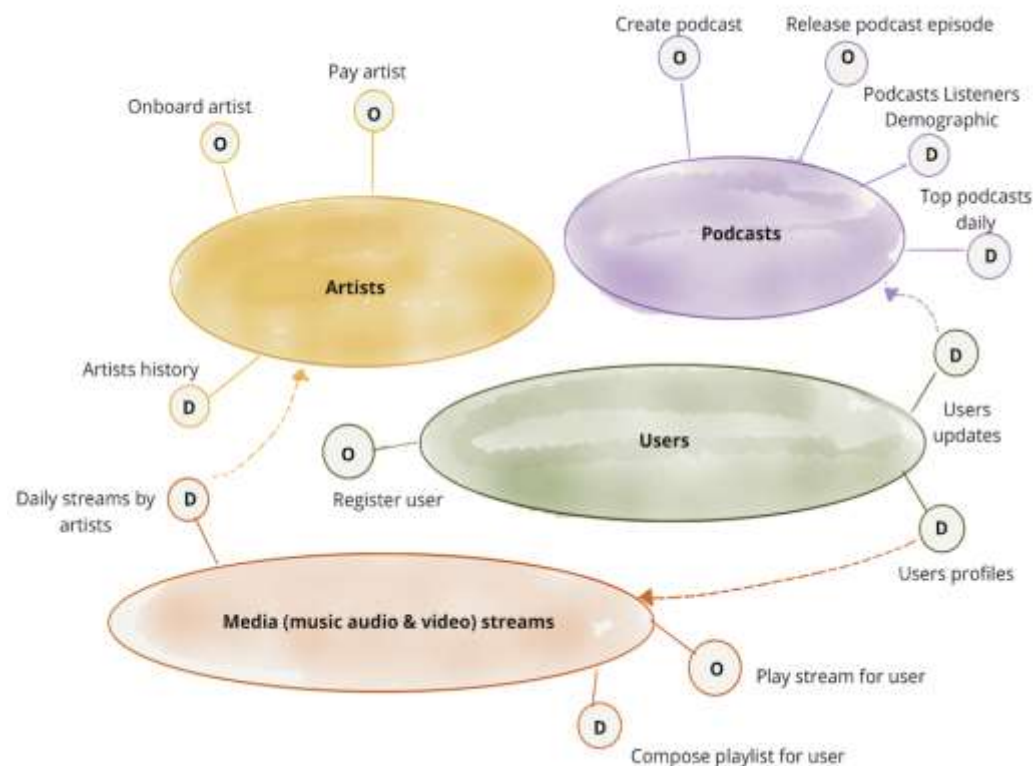
O que são domínios de dados

Produtos de dados

Plataforma de dados

Governança Federada

O que é um domínio



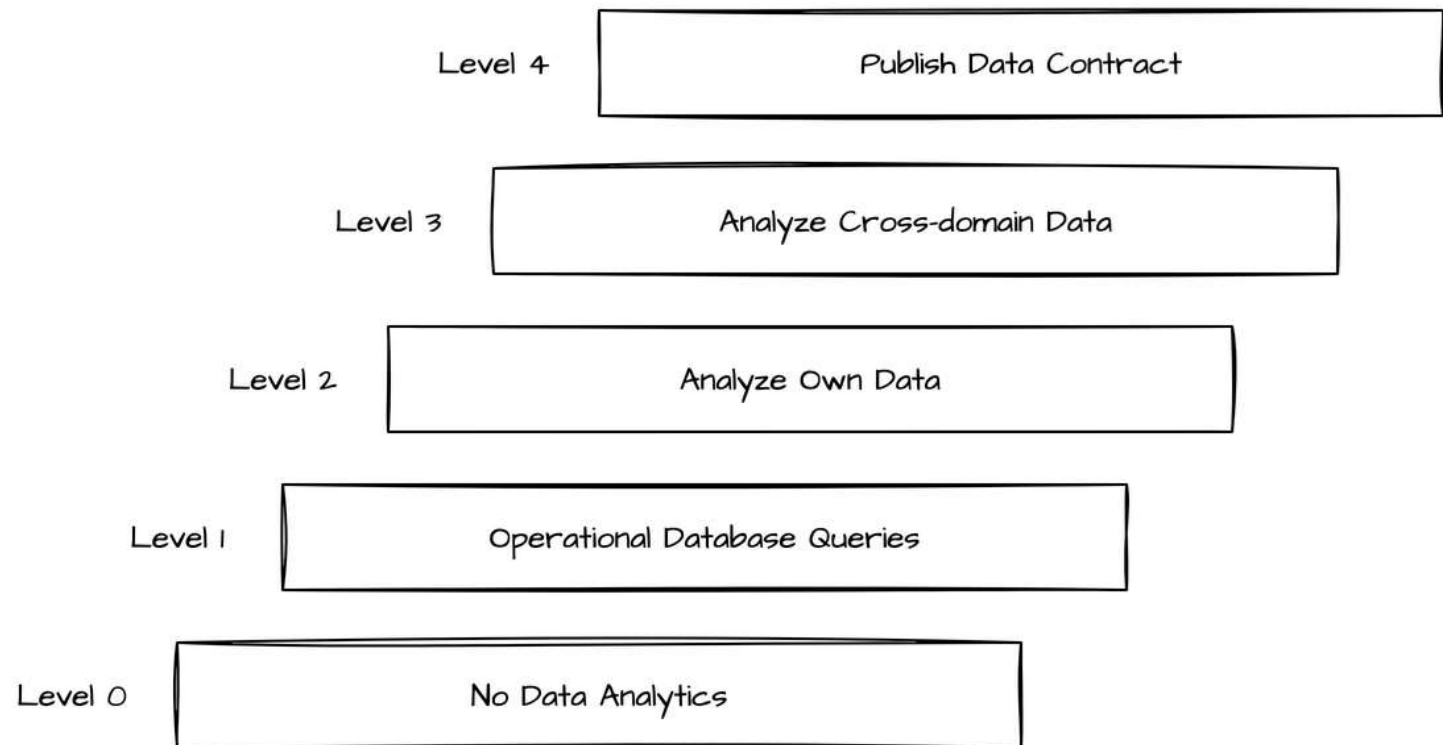
Domain Ownership

Data mesh, at core, is founded in *decentralization* and *distribution of responsibility* to people who are closest to the data in order to support continuous change and scalability. The question is, how do we decompose and decentralize the components of the data ecosystem and their ownership. The components here are made of *analytical data*, its *metadata*, and the *computation* necessary to serve it.

Data mesh follows the seams of organizational units as the axis of decomposition. Our organizations today are decomposed based on their business domains. Such decomposition localizes the impact of continuous change and evolution - for the most part - to the domain's bounded context. Hence, making the business domain's bounded context a good candidate for distribution of data ownership.

Domínio: Níveis de maturidade

Domain Team's Journey



datamesh-architecture.com

Data Mesh: Princípios

Domain-oriented decentralized data ownership and architecture	So that the ecosystem creating and consuming data can scale out as the number of sources of data, number of use cases, and diversity of access models to the data increases; simply increase the autonomous nodes on the mesh.
Data as a product	So that data users can easily discover, understand and securely use high quality data with a delightful experience; data that is distributed across many domains.
Self-serve data infrastructure as a platform	So that the domain teams can create and consume data products autonomously using the platform abstractions, hiding the complexity of building, executing and maintaining secure and interoperable data products.
Federated computational governance	So that data users can get value from aggregation and correlation of independent data products - the mesh is behaving as an ecosystem following global interoperability standards; standards that are baked computationally into the platform.

O que vamos abordar nesta aula

O que é Data Mesh

O que são domínios de dados

Produtos de dados

Plataforma de dados

Governança Federada

Produto de dados: O que é e como funciona

Data Product

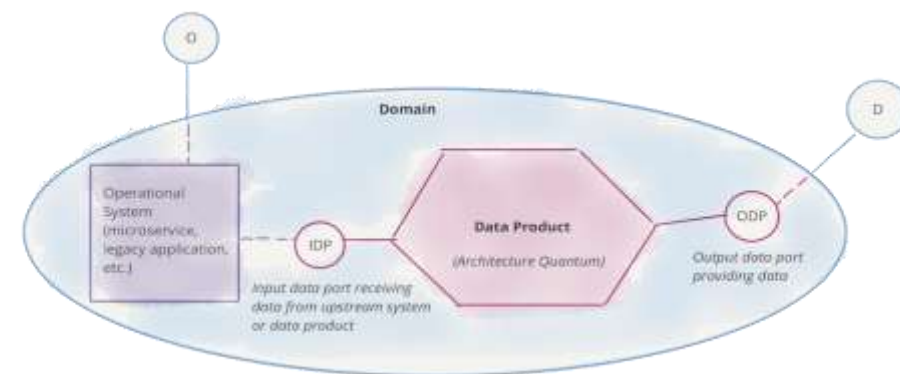
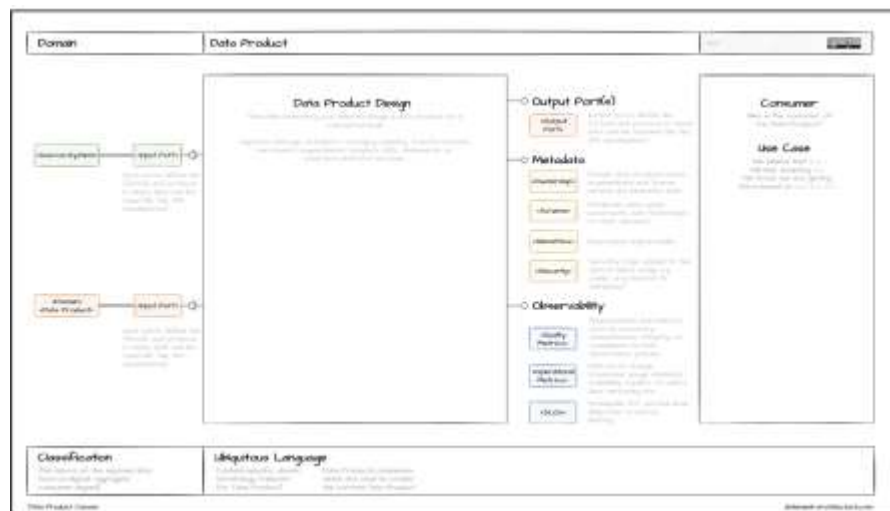
A data product is a logical unit that contains all components to process and store domain data for analytical or data-intensive use cases and makes them available to other teams via output ports. You can think of a microservice, but for analytical data.

<https://www.datamesh-architecture.com/>

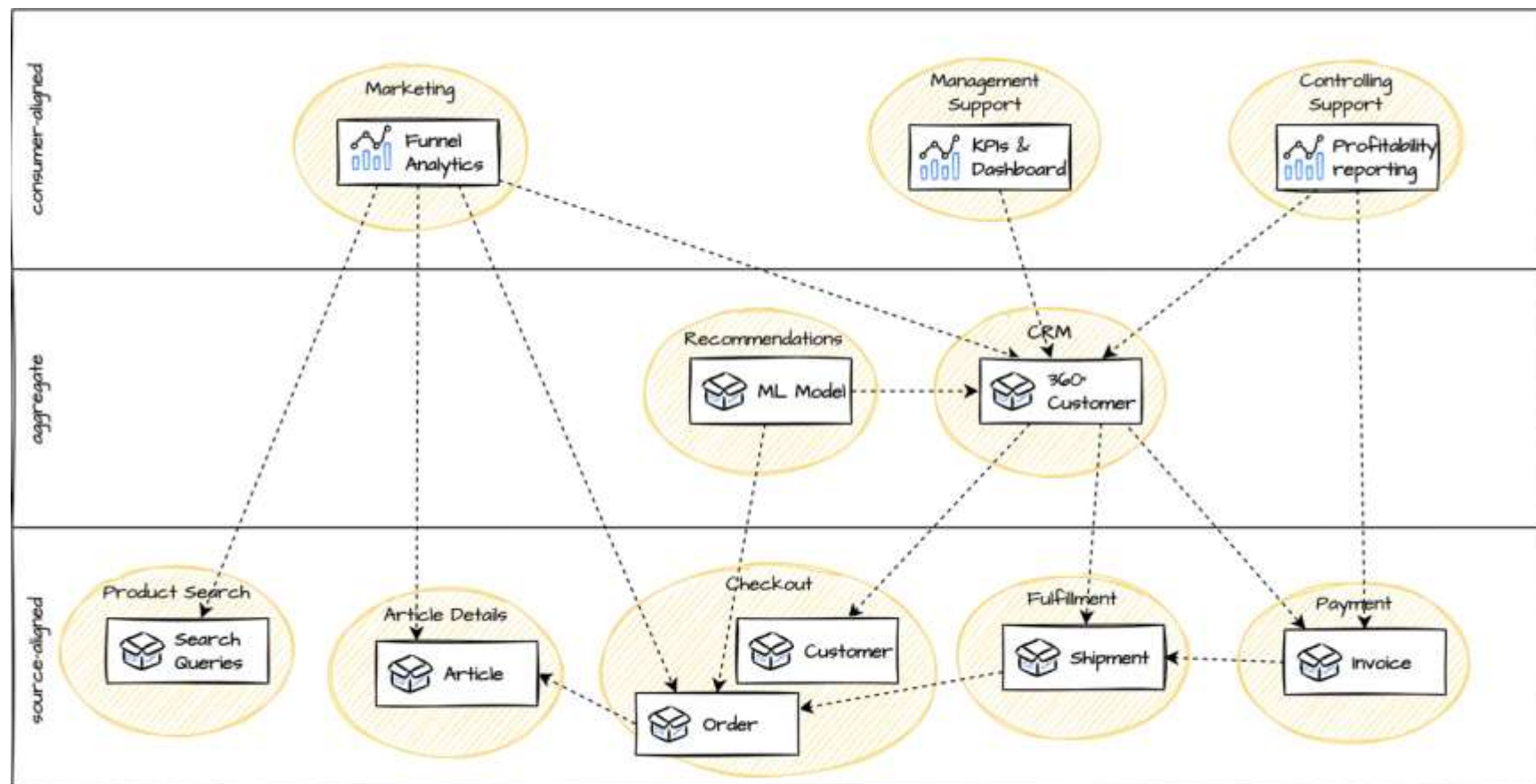
Logical architecture: data product the architectural quantum

Architecturally, to support data as a product that domains can autonomously serve or consume, data mesh introduces the concept of **data product** as its **architectural quantum**. Architectural quantum, as defined by Evolutionary Architecture, is the smallest unit of architecture that can be independently deployed with high functional cohesion, and includes all the structural elements required for its function.

<https://martinfowler.com/articles/data-mesh-principles.html>

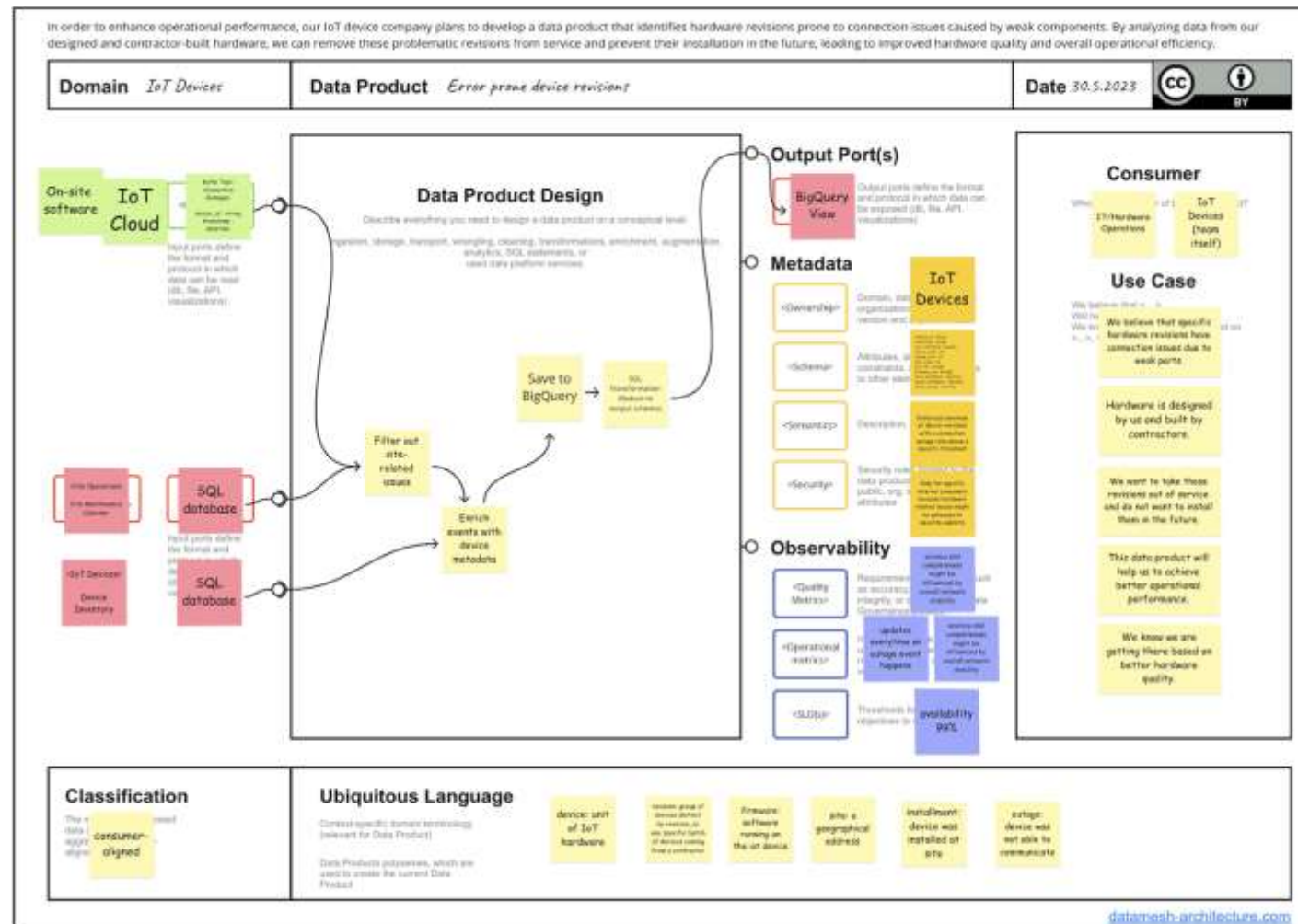


Tipos de Produto de dados



datamesh-architecture.com

Produto de dados: Data Product Canvas

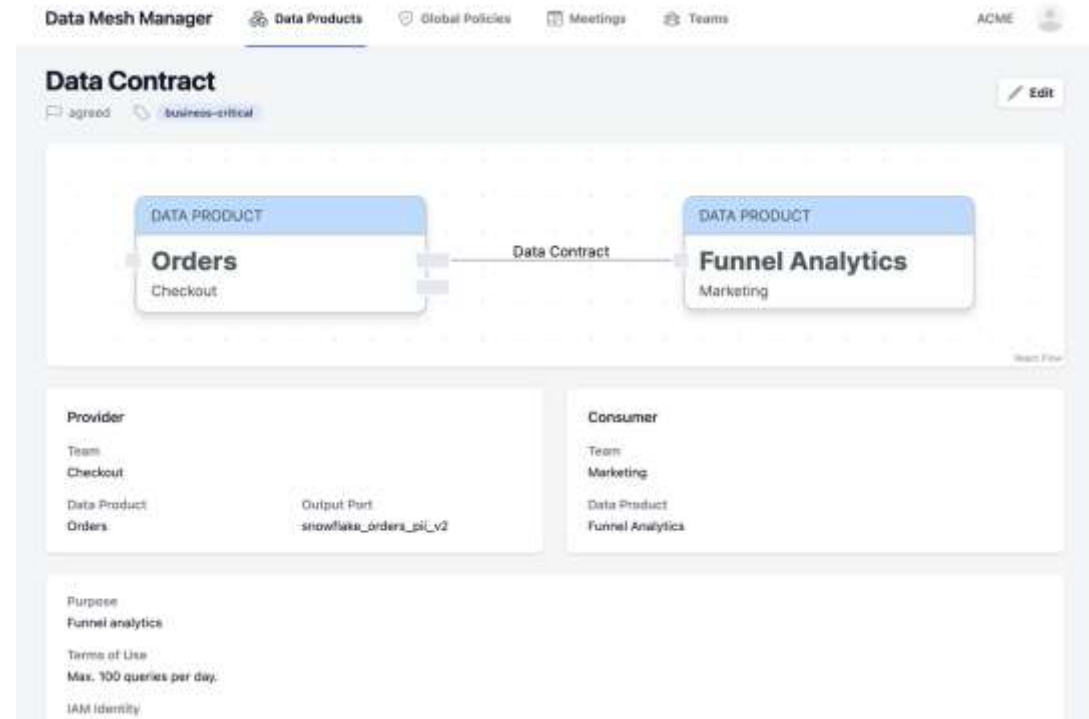


Produto de dados: Contratos de dados

Data Contract

A data contract is a formal agreement between two parties to use a data product. It specifies the guarantees about a provided data set and expectations concerning data product usage. It covers:

- Data Product Provider, including team, owner, and the output port to access
- Data Product Consumer, including team, and responsible contact
- Purpose of data usage
- Schema and semantics of used data attributes
- Service-level objectives, such as latency, availability, availability
- Terms, such as query intervals and data processing volumes
- Costs
- Start date
- End date
- Notice period, to cancel the contract



Data Mesh: Princípios

Domain-oriented decentralized data ownership and architecture	So that the ecosystem creating and consuming data can scale out as the number of sources of data, number of use cases, and diversity of access models to the data increases; simply increase the autonomous nodes on the mesh.
Data as a product	So that data users can easily discover, understand and securely use high quality data with a delightful experience; data that is distributed across many domains.
Self-serve data infrastructure as a platform	So that the domain teams can create and consume data products autonomously using the platform abstractions, hiding the complexity of building, executing and maintaining secure and interoperable data products.
Federated computational governance	So that data users can get value from aggregation and correlation of independent data products - the mesh is behaving as an ecosystem following global interoperability standards; standards that are baked computationally into the platform.

O que vamos abordar nesta aula

O que é Data Mesh

O que são domínios de dados

Produtos de dados

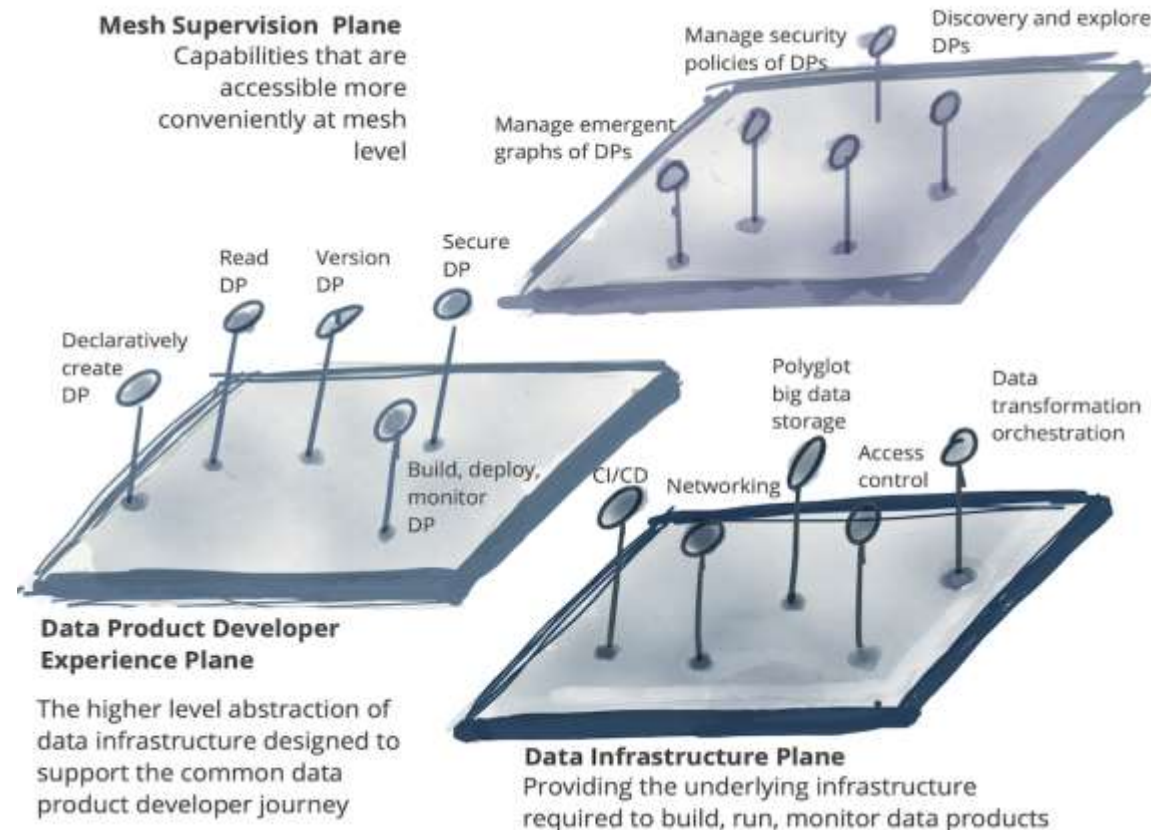
Plataforma de dados

Governança Federada

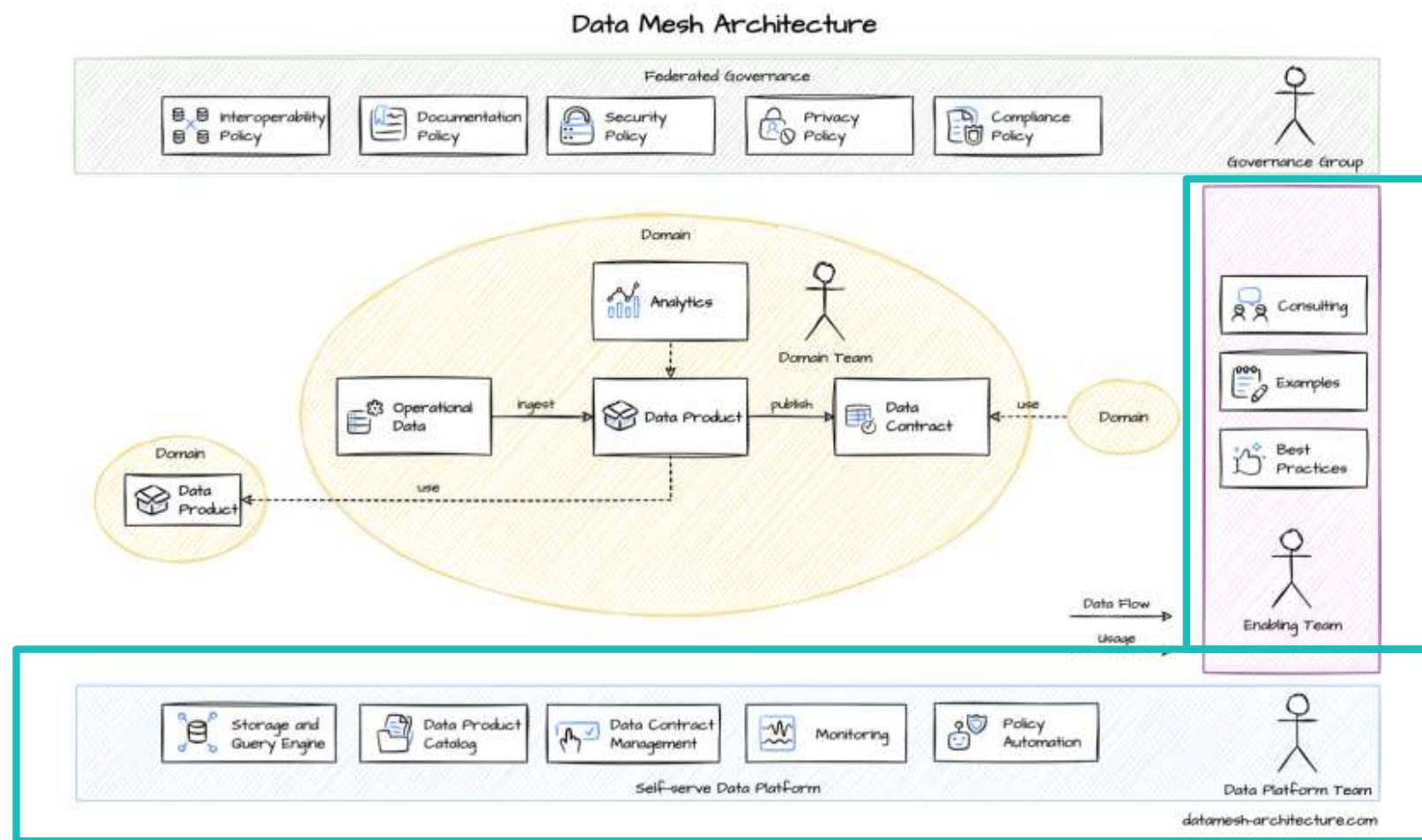
O que é a plataforma de dados

Self-serve data platform

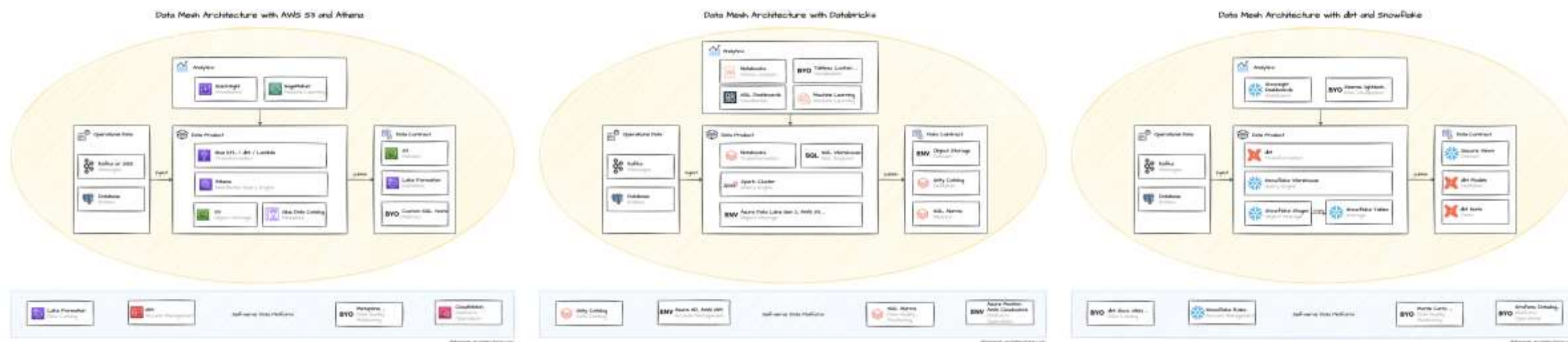
As you can imagine, to build, deploy, execute, monitor, and access a humble hexagon - a data product - there is a fair bit of infrastructure that needs to be provisioned and run; the skills needed to provision this infrastructure are specialized and would be difficult to replicate in each domain. Most importantly, the only way that teams can autonomously own their data products is to have access to a high-level abstraction of infrastructure that removes complexity and friction of provisioning and managing the lifecycle of data products. This calls for a new principle, *Self-serve data infrastructure as a platform to enable domain autonomy*.



O que é a plataforma de dados



Plataforma de dados: Stacks de plataforma



Data Mesh: Princípios

Domain-oriented
decentralized data
ownership and
architecture

So that the ecosystem creating and consuming data can scale out as the number of sources of data, number of use cases, and diversity of access models to the data increases; simply increase the autonomous nodes on the mesh.

Data as a product

So that data users can easily discover, understand and securely use high quality data with a delightful experience; data that is distributed across many domains.

Self-serve data
infrastructure as a
platform

So that the domain teams can create and consume data products autonomously using the platform abstractions, hiding the complexity of building, executing and maintaining secure and interoperable data products.

Federated
computational
governance

So that data users can get value from aggregation and correlation of independent data products - the mesh is behaving as an ecosystem following global interoperability standards; standards that are baked computationally into the platform.

O que vamos abordar nesta aula

O que é Data Mesh

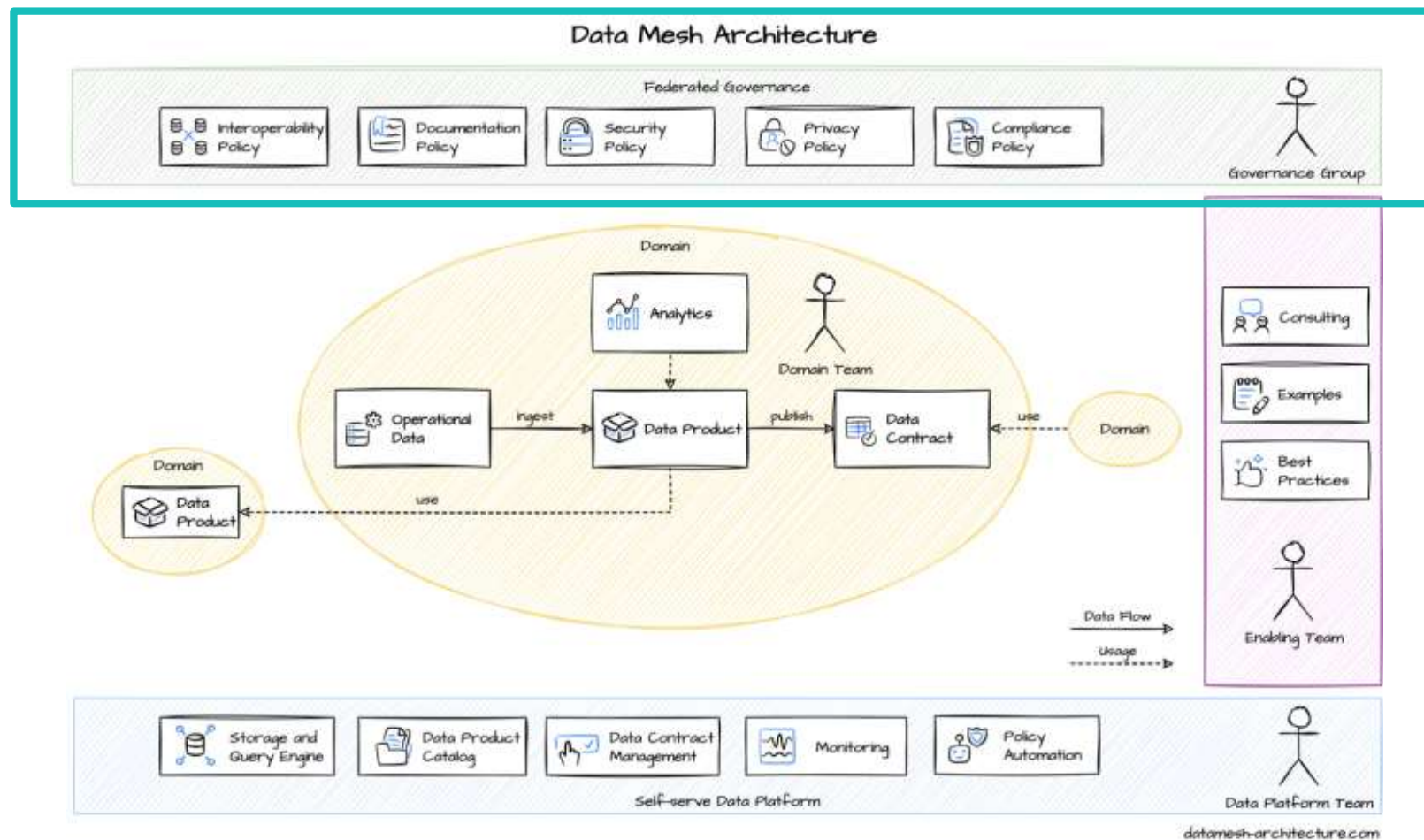
O que são domínios de dados

Produtos de dados

Plataforma de dados

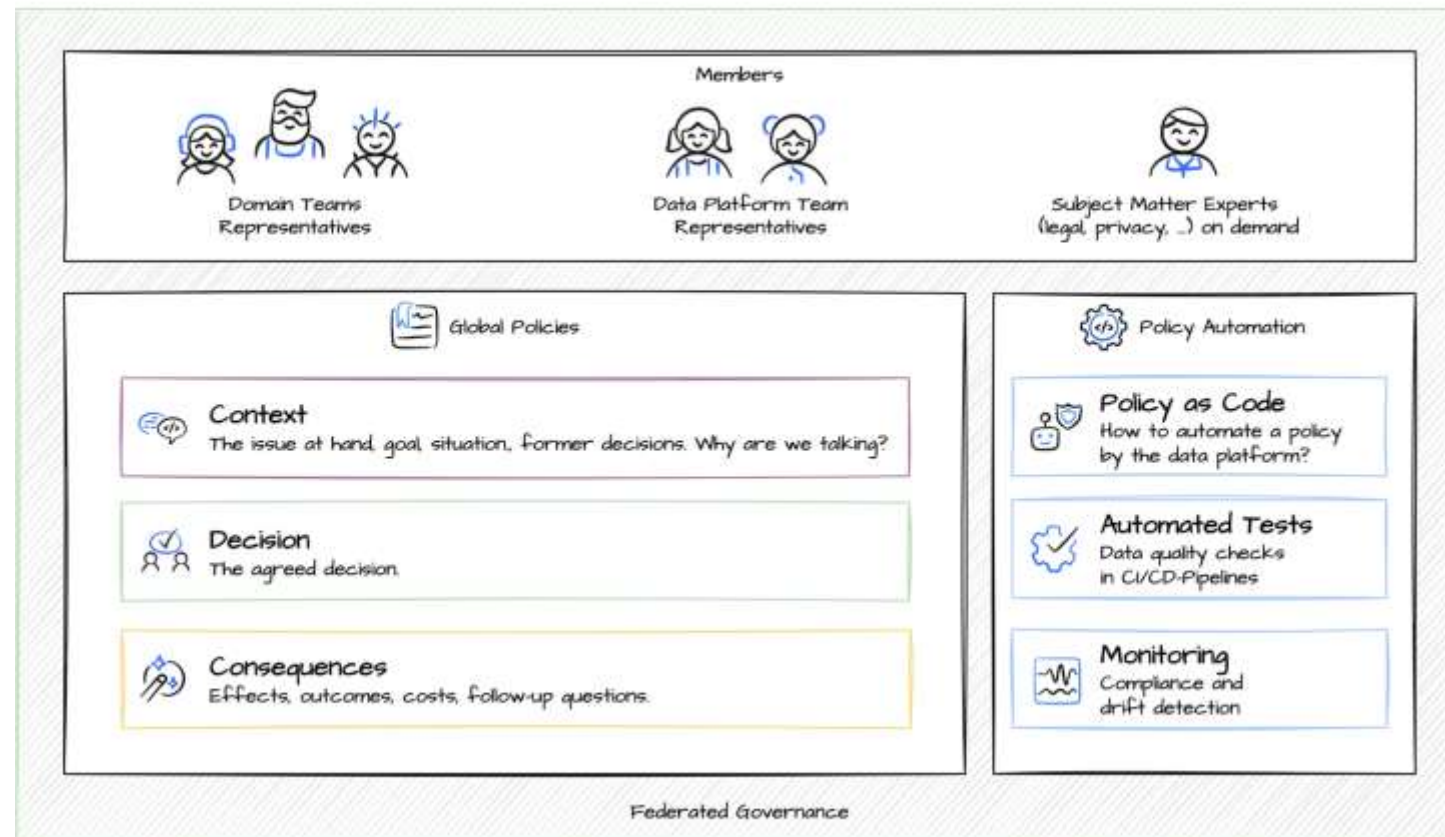
Governança Federada

Governança Federada de dados



Governança Federada de dados

Data Mesh Governance



datamesh-governance.com

Governança Federada de dados

Pre data mesh governance aspect	Data mesh governance aspect
Centralized team	Federated team
Responsible for data quality	Responsible for defining how to model what constitutes quality
Responsible for data security	Responsible for defining aspects of data security i.e. data sensitivity levels for the platform to build in and monitor automatically
Responsible for complying with regulation	Responsible for defining the regulation requirements for the platform to build in and monitor automatically
Centralized custodianship of data	Federated custodianship of data by domains
Responsible for global canonical data modeling	Responsible for modeling polysems - data elements that cross the boundaries of multiple domains
Team is independent from domains	Team is made of domains representatives
Aiming for a well defined static structure of data	Aiming for enabling effective mesh operation embracing a continuously changing and a dynamic topology of the mesh
Centralized technology used by monolithic lake/warehouse	Self-serve platform technologies used by each domain
Measure success based on number or volume of governed data (tables)	Measure success based on the network effect - the connections representing the consumption of data on the mesh
Manual process with human intervention	Automated processes implemented by the platform
Prevent error	Detect error and recover through platform's automated processing



PUC Minas